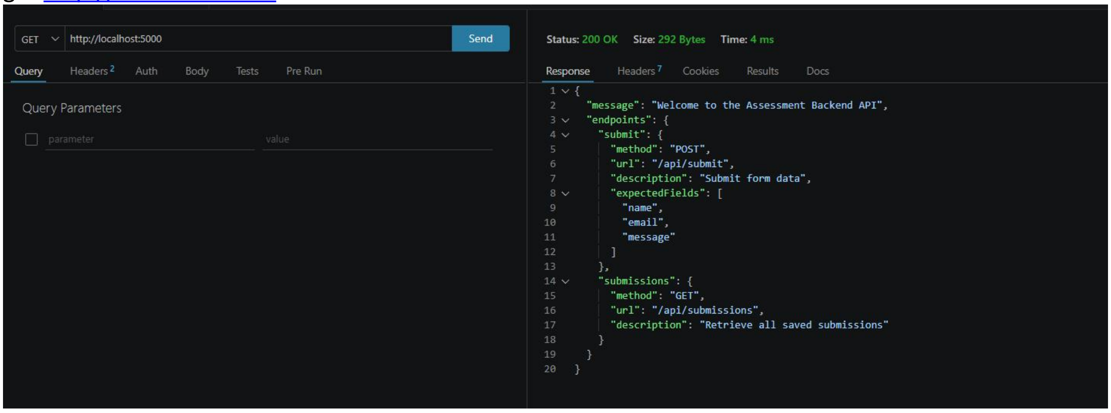
Huawei Technical Test Result Detail

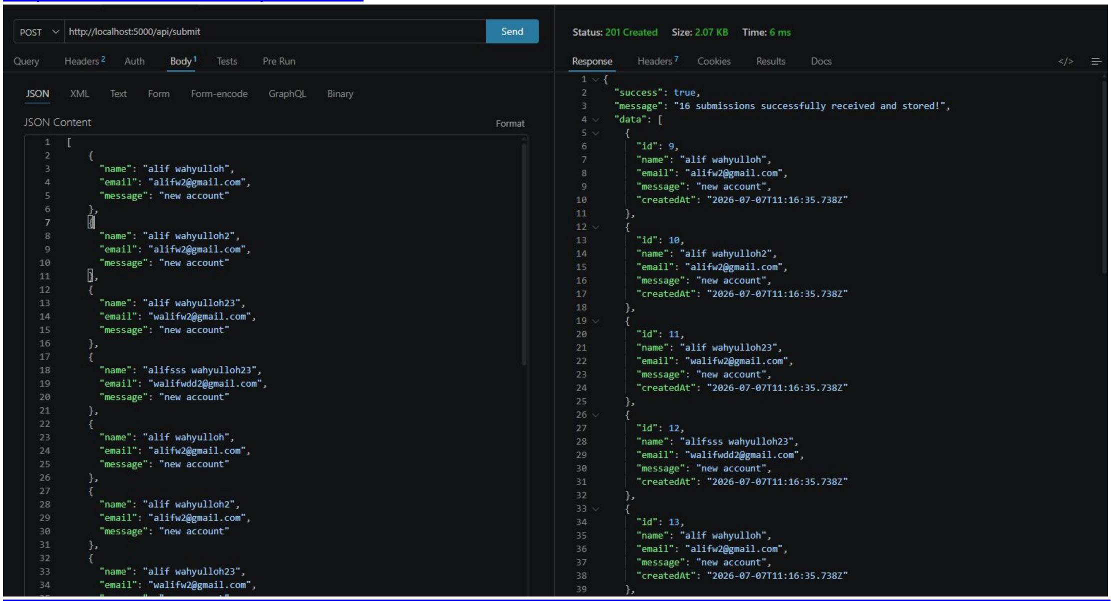
Task 1 Backend

get http://localhost:5000

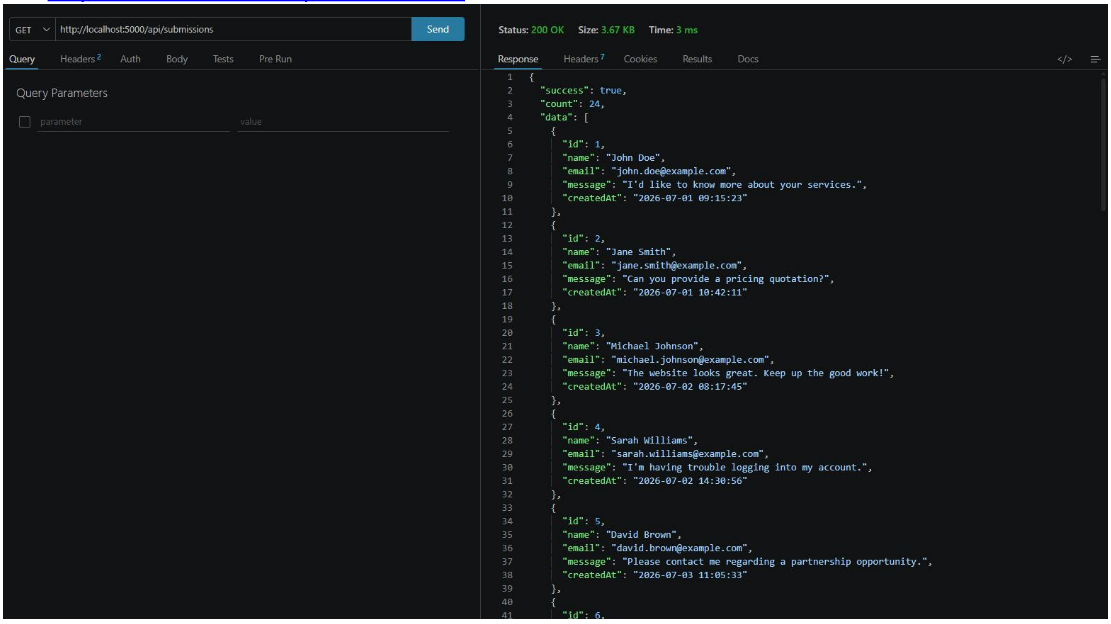


Post

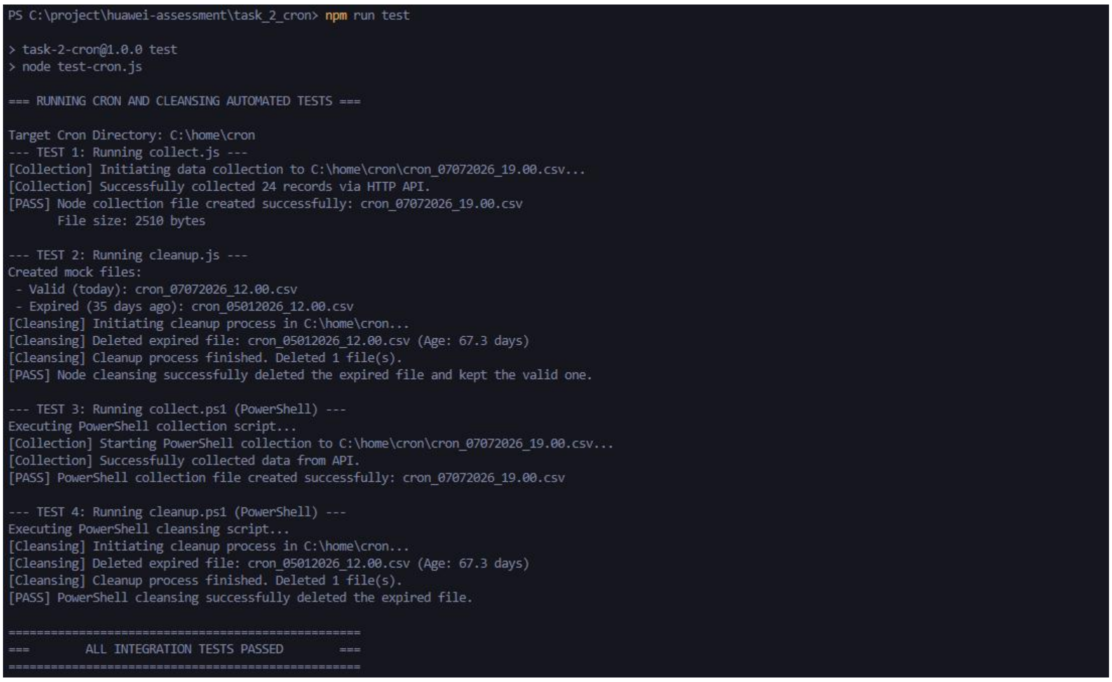
http://localhost:5000/api/submit



Get http://localhost:5000/api/submissions

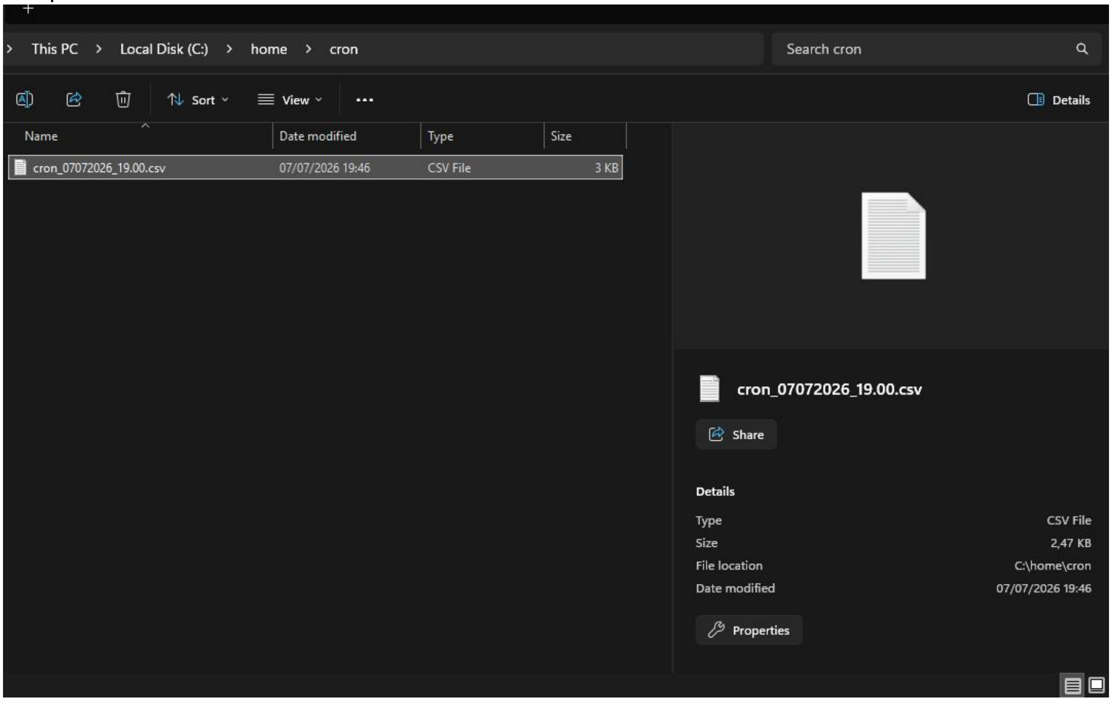


Task 2 Cron Job



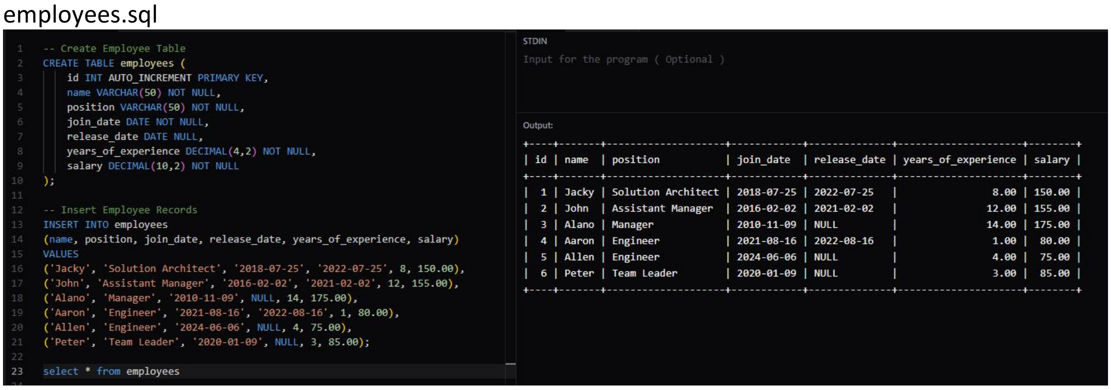
```
PS C:\project\huawei-assessment\task_2_cron> npm run test

> task-2-cron@1.0.0 test
> node test-cron.js

=== RUNNING CRON AND CLEANSING AUTOMATED TESTS ===

Target Cron Directory: C:\home\cron
--- TEST 1: Running collect.js ---
[Collection] Initiating data collection to C:\home\cron\cron_07072026_19.00.csv...
[Collection] Successfully collected 24 records via HTTP API.
[PASS] Node collection file created successfully: cron_07072026_19.00.csv
       File size: 2510 bytes

--- TEST 2: Running cleanup.js ---
Created mock files:
 - Valid (today): cron_07072026_12.00.csv
 - Expired (35 days ago): cron_05012026_12.00.csv
[Cleansing] Initiating cleanup process in C:\home\cron...
[Cleansing] Deleted expired file: cron_05012026_12.00.csv (Age: 67.3 days)
[Cleansing] Cleanup process finished. Deleted 1 file(s).
[PASS] Node cleansing successfully deleted the expired file and kept the valid one.

--- TEST 3: Running collect.ps1 (PowerShell) ---
Executing PowerShell collection script...
[Collection] Starting PowerShell collection to C:\home\cron\cron_07072026_19.00.csv...
[Collection] Successfully collected data from API.
[PASS] PowerShell collection file created successfully: cron_07072026_19.00.csv

--- TEST 4: Running cleanup.ps1 (PowerShell) ---
Executing PowerShell cleansing script...
[Cleansing] Initiating cleanup process in C:\home\cron...
[Cleansing] Deleted expired file: cron_05012026_12.00.csv (Age: 67.3 days)
[Cleansing] Cleanup process finished. Deleted 1 file(s).
[PASS] PowerShell cleansing successfully deleted the expired file.

==================================================
===        ALL INTEGRATION TESTS PASSED        ===
==================================================
```

Sample folder & file created



Task 3 SQL Query

employees.sql



```sql
-- Create Employee Table
CREATE TABLE employees (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    position VARCHAR(50) NOT NULL,
    join_date DATE NOT NULL,
    release_date DATE NULL,
    years_of_experience DECIMAL(4,2) NOT NULL,
    salary DECIMAL(10,2) NOT NULL
);

-- Insert Employee Records
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Jacky', 'Solution Architect', '2018-07-25', '2022-07-25', 8, 150.00),
('John', 'Assistant Manager', '2016-02-02', '2021-02-02', 12, 155.00),
('Alano', 'Manager', '2010-11-09', NULL, 14, 175.00),
('Aaron', 'Engineer', '2021-08-16', '2022-08-16', 1, 80.00),
('Allen', 'Engineer', '2024-06-06', NULL, 4, 75.00),
('Peter', 'Team Leader', '2020-01-09', NULL, 3, 85.00);

select * from employees
```

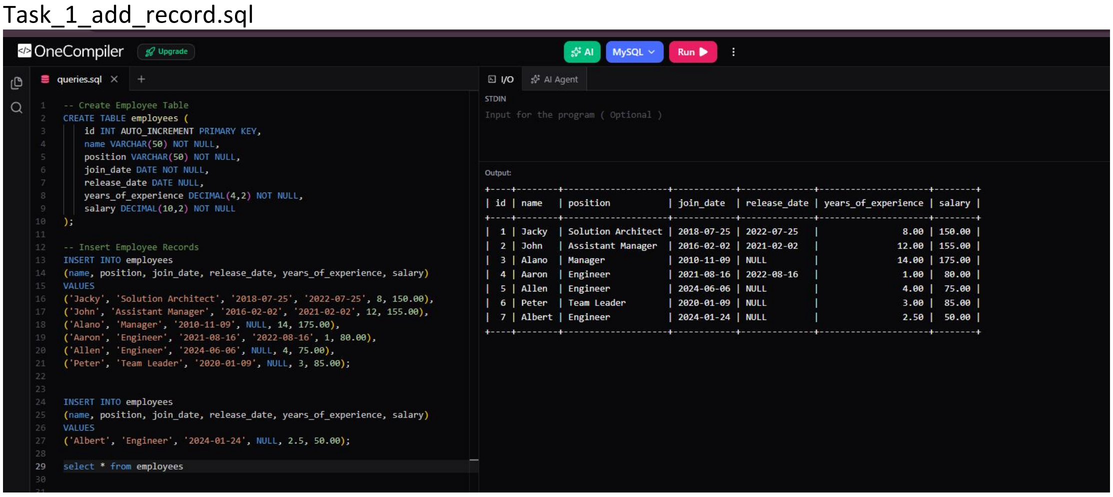
STDIN

Input for the program (Optional)

Output:

id	name	position	join_date	release_date	years_of_experience	salary
1	Jacky	Solution Architect	2018-07-25	2022-07-25	8.00	150.00
2	John	Assistant Manager	2016-02-02	2021-02-02	12.00	155.00
3	Alano	Manager	2010-11-09	NULL	14.00	175.00
4	Aaron	Engineer	2021-08-16	2022-08-16	1.00	80.00
5	Allen	Engineer	2024-06-06	NULL	4.00	75.00
6	Peter	Team Leader	2020-01-09	NULL	3.00	85.00

Task_1_add_record.sql




```sql
-- Create Employee Table
CREATE TABLE employees (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    position VARCHAR(50) NOT NULL,
    join_date DATE NOT NULL,
    release_date DATE NULL,
    years_of_experience DECIMAL(4,2) NOT NULL,
    salary DECIMAL(10,2) NOT NULL
);

-- Insert Employee Records
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Jacky', 'Solution Architect', '2018-07-25', '2022-07-25', 8, 150.00),
('John', 'Assistant Manager', '2016-02-02', '2021-02-02', 12, 155.00),
('Alano', 'Manager', '2010-11-09', NULL, 14, 175.00),
('Aaron', 'Engineer', '2021-08-16', '2022-08-16', 1, 80.00),
('Allen', 'Engineer', '2024-06-06', NULL, 4, 75.00),
('Peter', 'Team Leader', '2020-01-09', NULL, 3, 85.00);


INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Albert', 'Engineer', '2024-01-24', NULL, 2.5, 50.00);

select * from employees
```

STDIN

Input for the program (Optional)

Output:

id	name	position	join_date	release_date	years_of_experience	salary
1	Jacky	Solution Architect	2018-07-25	2022-07-25	8.00	150.00
2	John	Assistant Manager	2016-02-02	2021-02-02	12.00	155.00
3	Alano	Manager	2010-11-09	NULL	14.00	175.00
4	Aaron	Engineer	2021-08-16	2022-08-16	1.00	80.00
5	Allen	Engineer	2024-06-06	NULL	4.00	75.00
6	Peter	Team Leader	2020-01-09	NULL	3.00	85.00
7	Albert	Engineer	2024-01-24	NULL	2.50	50.00

Task_2_update_engineer_salary.sql

```sql
-- Create Employee Table
CREATE TABLE employees (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    position VARCHAR(50) NOT NULL,
    join_date DATE NOT NULL,
    release_date DATE NULL,
    years_of_experience DECIMAL(4,2) NOT NULL,
    salary DECIMAL(10,2) NOT NULL
);

-- Insert Employee Records
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Jacky', 'Solution Architect', '2018-07-25', '2022-07-25', 8, 150.00),
('John', 'Assistant Manager', '2016-02-02', '2021-02-02', 12, 155.00),
('Alano', 'Manager', '2010-11-09', NULL, 14, 175.00),
('Aaron', 'Engineer', '2021-08-16', '2022-08-16', 1, 80.00),
('Allen', 'Engineer', '2024-06-06', NULL, 4, 75.00),
('Peter', 'Team Leader', '2020-01-09', NULL, 3, 85.00);

INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Albert', 'Engineer', '2024-01-24', NULL, 2.5, 50.00);

-- Update salary for employees with position 'Engineer' to 85.00
UPDATE employees
SET salary = 85.00
WHERE position = 'Engineer';

select * from employees
```

Output:

id	name	position	join_date	release_date	years_of_experience	salary
1	Jacky	Solution Architect	2018-07-25	2022-07-25	8.00	150.00
2	John	Assistant Manager	2016-02-02	2021-02-02	12.00	155.00
3	Alano	Manager	2010-11-09	NULL	14.00	175.00
4	Aaron	Engineer	2021-08-16	2022-08-16	1.00	85.00
5	Allen	Engineer	2024-06-06	NULL	4.00	85.00
6	Peter	Team Leader	2020-01-09	NULL	3.00	85.00
7	Albert	Engineer	2024-01-24	NULL	2.50	85.00

Task_3_total_salary_2021.sql

```sql
-- Create Employee Table
CREATE TABLE employees (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    position VARCHAR(50) NOT NULL,
    join_date DATE NOT NULL,
    release_date DATE NULL,
    years_of_experience DECIMAL(4,2) NOT NULL,
    salary DECIMAL(10,2) NOT NULL
);

-- Insert Employee Records
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Jacky', 'Solution Architect', '2018-07-25', '2022-07-25', 8, 150.00),
('John', 'Assistant Manager', '2016-02-02', '2021-02-02', 12, 155.00),
('Alano', 'Manager', '2010-11-09', NULL, 14, 175.00),
('Aaron', 'Engineer', '2021-08-16', '2022-08-16', 1, 80.00),
('Allen', 'Engineer', '2024-06-06', NULL, 4, 75.00),
('Peter', 'Team Leader', '2020-01-09', NULL, 3, 85.00);
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Albert', 'Engineer', '2024-01-24', NULL, 2.5, 50.00);
-- Update salary for employees with position 'Engineer' to 85.00
UPDATE employees
SET salary = 85.00
WHERE position = 'Engineer';
select * from employees;

-- Hitung total pengeluaran salary saat tahun 2021.
SELECT SUM(salary) AS 'total salary 2021'
FROM employees
WHERE join_date <= '2021-12-31'
  AND (release_date IS NULL OR release_date >= '2021-01-01');
```

Output:

id	name	position	join_date	release_date	years_of_experience	salary
1	Jacky	Solution Architect	2018-07-25	2022-07-25	8.00	150.00
2	John	Assistant Manager	2016-02-02	2021-02-02	12.00	155.00
3	Alano	Manager	2010-11-09	NULL	14.00	175.00
4	Aaron	Engineer	2021-08-16	2022-08-16	1.00	85.00
5	Allen	Engineer	2024-06-06	NULL	4.00	85.00
6	Peter	Team Leader	2020-01-09	NULL	3.00	85.00
7	Albert	Engineer	2024-01-24	NULL	2.50	85.00

total salary 2021
650.00

Task_4_top_experiece.sql

OneCompiler Upgrade AI MySQL Run

queries.sql

```sql
-- Create Employee Table
CREATE TABLE employees (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    position VARCHAR(50) NOT NULL,
    join_date DATE NOT NULL,
    release_date DATE NULL,
    years_of_experience DECIMAL(4,2) NOT NULL,
    salary DECIMAL(10,2) NOT NULL
);

-- Insert Employee Records
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Jacky', 'Solution Architect', '2018-07-25', '2022-07-25', 8, 150.00),
('John', 'Assistant Manager', '2016-02-02', '2021-02-02', 12, 155.00),
('Alano', 'Manager', '2010-11-09', NULL, 14, 175.00),
('Aaron', 'Engineer', '2021-08-16', '2022-08-16', 1, 80.00),
('Allen', 'Engineer', '2024-06-06', NULL, 4, 75.00),
('Peter', 'Team Leader', '2020-01-09', NULL, 3, 85.00);
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Albert', 'Engineer', '2024-01-24', NULL, 2.5, 50.00);
-- Update salary for employees with position 'Engineer' to 85.00
UPDATE employees
SET salary = 85.00
WHERE position = 'Engineer';

select * from employees;
-- Menampilkan 3 employee paling banyak yang memiliki Years of Experience
SELECT *
FROM employees
ORDER BY years_of_experience DESC
LIMIT 3;
```

I/O AI Agent

STDIN

Input for the program (Optional)

Output:

```
+----+--------+--------------------+------------+--------------+---------------------+--------+
| id | name   | position           | join_date  | release_date | years_of_experience | salary |
+----+--------+--------------------+------------+--------------+---------------------+--------+
|  1 | Jacky  | Solution Architect | 2018-07-25 | 2022-07-25   |                8.00 | 150.00 |
|  2 | John   | Assistant Manager  | 2016-02-02 | 2021-02-02   |               12.00 | 155.00 |
|  3 | Alano  | Manager            | 2010-11-09 | NULL         |               14.00 | 175.00 |
|  4 | Aaron  | Engineer           | 2021-08-16 | 2022-08-16   |                1.00 |  85.00 |
|  5 | Allen  | Engineer           | 2024-06-06 | NULL         |                4.00 |  85.00 |
|  6 | Peter  | Team Leader        | 2020-01-09 | NULL         |                3.00 |  85.00 |
|  7 | Albert | Engineer           | 2024-01-24 | NULL         |                2.50 |  85.00 |
+----+--------+--------------------+------------+--------------+---------------------+--------+
+----+-------+--------------------+------------+--------------+---------------------+--------+
| id | name  | position           | join_date  | release_date | years_of_experience | salary |
+----+-------+--------------------+------------+--------------+---------------------+--------+
|  3 | Alano | Manager            | 2010-11-09 | NULL         |               14.00 | 175.00 |
|  2 | John  | Assistant Manager  | 2016-02-02 | 2021-02-02   |               12.00 | 155.00 |
|  1 | Jacky | Solution Architect | 2018-07-25 | 2022-07-25   |                8.00 | 150.00 |
+----+-------+--------------------+------------+--------------+---------------------+--------+
```

Task_5_sub_query_engineer.sql

queries.sql

```sql
-- Create Employee Table
CREATE TABLE employees (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    position VARCHAR(50) NOT NULL,
    join_date DATE NOT NULL,
    release_date DATE NULL,
    years_of_experience DECIMAL(4,2) NOT NULL,
    salary DECIMAL(10,2) NOT NULL
);

-- Insert Employee Records
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Jacky', 'Solution Architect', '2018-07-25', '2022-07-25', 8, 150.00),
('John', 'Assistant Manager', '2016-02-02', '2021-02-02', 12, 155.00),
('Alano', 'Manager', '2010-11-09', NULL, 14, 175.00),
('Aaron', 'Engineer', '2021-08-16', '2022-08-16', 1, 80.00),
('Allen', 'Engineer', '2024-06-06', NULL, 4, 75.00),
('Peter', 'Team Leader', '2020-01-09', NULL, 3, 85.00);
INSERT INTO employees
(name, position, join_date, release_date, years_of_experience, salary)
VALUES
('Albert', 'Engineer', '2024-01-24', NULL, 2.5, 50.00);
-- Update salary for employees with position 'Engineer' to 85.00
UPDATE employees
SET salary = 85.00
WHERE position = 'Engineer';
select * from employees;

-- Subquery untuk employee dengan posisi Engineer yang memiliki experience <= 3
SELECT *
FROM employees
WHERE id IN (
    SELECT id
    FROM employees
    WHERE position = 'Engineer'
      AND years_of_experience <= 3
);
```

I/O AI Agent

STDIN

Input for the program (Optional)

Output:

```
+----+--------+--------------------+------------+--------------+---------------------+--------+
| id | name   | position           | join_date  | release_date | years_of_experience | salary |
+----+--------+--------------------+------------+--------------+---------------------+--------+
|  1 | Jacky  | Solution Architect | 2018-07-25 | 2022-07-25   |                8.00 | 150.00 |
|  2 | John   | Assistant Manager  | 2016-02-02 | 2021-02-02   |               12.00 | 155.00 |
|  3 | Alano  | Manager            | 2010-11-09 | NULL         |               14.00 | 175.00 |
|  4 | Aaron  | Engineer           | 2021-08-16 | 2022-08-16   |                1.00 |  85.00 |
|  5 | Allen  | Engineer           | 2024-06-06 | NULL         |                4.00 |  85.00 |
|  6 | Peter  | Team Leader        | 2020-01-09 | NULL         |                3.00 |  85.00 |
|  7 | Albert | Engineer           | 2024-01-24 | NULL         |                2.50 |  85.00 |
+----+--------+--------------------+------------+--------------+---------------------+--------+
+----+--------+----------+------------+--------------+---------------------+--------+
| id | name   | position | join_date  | release_date | years_of_experience | salary |
+----+--------+----------+------------+--------------+---------------------+--------+
|  4 | Aaron  | Engineer | 2021-08-16 | 2022-08-16   |                1.00 |  85.00 |
|  7 | Albert | Engineer | 2024-01-24 | NULL         |                2.50 |  85.00 |
+----+--------+----------+------------+--------------+---------------------+--------+
```